

Session 3 — Linear Algebra I

Solution sheet

Complete worked solutions, with reminders of the key ideas. Numerical answers are boxed. Emphasise the geometry: dot products measure similarity, and matrices act on space.

Exercise 1 $\mathbf{u} = (2, -1)$, $\mathbf{v} = (1, 3)$.

Solution.

(a) $3\mathbf{u} = (6, -3)$, $2\mathbf{v} = (2, 6)$, so $3\mathbf{u} - 2\mathbf{v} = \boxed{(4, -9)}$.

(b) $\|\mathbf{u}\| = \sqrt{2^2 + (-1)^2} = \sqrt{5} = \boxed{\approx 2.236}$.

(c) Unit vector $\frac{\mathbf{u}}{\|\mathbf{u}\|} = \frac{(2, -1)}{\sqrt{5}} \approx \boxed{(0.894, -0.447)}$.

Reminder

A unit vector keeps only *direction* (length 1). Dividing any nonzero vector by its norm normalises it.

Exercise 2 Write $(5, 1) = a(1, 1) + b(1, -1)$.

Solution. Matching components: $a + b = 5$ and $a - b = 1$. Adding gives $2a = 6 \Rightarrow a = 3$, then $b = 2$. So $(5, 1) = \boxed{3(1, 1) + 2(1, -1)}$. Because $(1, 1)$ and $(1, -1)$ are independent, they *span the whole plane*: every vector is some combination of them.

Reminder

Two linearly independent vectors in $\mathbb{R}^2$ form a *basis*: every vector has a unique representation as a linear combination of them.

Exercise 3 $\mathbf{a} = (3, 4, 0)$, $\mathbf{b} = (0, 4, 3)$.

Solution.

(a) $\mathbf{a} \cdot \mathbf{b} = 0 + 16 + 0 = 16$; $\|\mathbf{a}\| = \sqrt{9 + 16} = 5$, $\|\mathbf{b}\| = \sqrt{16 + 9} = 5$.

(b) $\cos \theta = \frac{16}{5 \cdot 5} = \boxed{0.64}$, so $\theta = \arccos(0.64) \approx \boxed{50.2^\circ}$.

(c) A high cosine similarity means the two stimuli evoke *similar* population activity patterns.

Reminder

Cosine similarity = $\frac{\mathbf{a} \cdot \mathbf{b}}{\|\mathbf{a}\| \|\mathbf{b}\|}$ measures alignment regardless of magnitude — the standard tool for comparing neural representations or word embeddings.

Exercise 4 $\mathbf{p} = (1, 2, 2)$, $\mathbf{q} = (3, 2, 4)$.

Solution. $\mathbf{q} - \mathbf{p} = (2, 0, 2)$, so the distance is $\|\mathbf{q} - \mathbf{p}\| = \sqrt{2^2 + 0^2 + 2^2} = \sqrt{8} = \boxed{\approx 2.83}$.

Exercise 5 $\mathbf{a} = (2, 1, -1)$, $\mathbf{b} = (1, -1, 1)$.

Solution.

(a) $\mathbf{a} \cdot \mathbf{b} = 2 - 1 - 1 = \boxed{0}$, so the vectors *are* orthogonal.

(b) Orthogonal population vectors mean the two stimuli are encoded along *independent, non-overlapping* directions of the neural space — their representations do not interfere.

Reminder

Dot product zero $\Leftrightarrow$ perpendicular. Orthogonality is the geometric version of “carrying no shared component”.

Exercise 6 Project $\mathbf{b} = (4, 2)$ onto $\mathbf{a} = (3, 0)$.

Solution. $\mathbf{a} \cdot \mathbf{b} = 12$, $\mathbf{a} \cdot \mathbf{a} = 9$, so

(a) $\text{proj}_{\mathbf{a}} \mathbf{b} = \frac{12}{9}(3, 0) = \boxed{(4, 0)}$.

(b) Residual = $\mathbf{b} - \text{proj}_{\mathbf{a}} \mathbf{b} = (0, 2)$, and $(0, 2) \cdot (3, 0) = 0$: orthogonal to $\mathbf{a}$, as it must be.

Reminder

The projection is the “shadow” of $\mathbf{b}$ along $\mathbf{a}$; the residual is what $\mathbf{a}$ cannot explain. Fitting a regression line *is* this projection.

Exercise 7 $\mathbf{W} = \begin{pmatrix} 2 & -1 \\ 1 & 3 \end{pmatrix}$, $\mathbf{x} = (1, 2)$.

Solution. $\mathbf{W}\mathbf{x} = (2(1) + (-1)(2), 1(1) + 3(2)) = \boxed{(0, 7)}$.

Reminder

Each output entry is the dot product of a *row* of $\mathbf{W}$ with $\mathbf{x}$; equivalently, $\mathbf{W}\mathbf{x}$ is a combination of the *columns* of $\mathbf{W}$.

Exercise 8 $\mathbf{A} = \begin{pmatrix} 1 & 2 \\ 0 & 1 \end{pmatrix}$, $\mathbf{B} = \begin{pmatrix} 2 & 0 \\ 0 & 3 \end{pmatrix}$.

Solution.

(a) $AB = \begin{pmatrix} 2 & 6 \\ 0 & 3 \end{pmatrix}$.

(b) $BA = \begin{pmatrix} 2 & 4 \\ 0 & 3 \end{pmatrix}$.

(c) They are *not* equal — “shear then scale” differs from “scale then shear”.

Reminder

Matrix multiplication composes transformations and is *not commutative*: AB means “do B , then A ”. Order matters.

Exercise 9 $A = \begin{pmatrix} 3 & 1 \\ 2 & 4 \end{pmatrix}$.

Solution.

(a) $\det A = 3(4) - 1(2) = 10$.

(b) A scales areas by $|\det| = 10$.

(c) Invertible, because $\det A \neq 0$.

Reminder

The determinant is the (signed) area-scaling factor. $\det = 0$ would mean the map collapses the plane onto a line — not invertible.

Exercise 10 Invert $A = \begin{pmatrix} 4 & 3 \\ 5 & 4 \end{pmatrix}$.

Solution. $\det A = 4(4) - 3(5) = 16 - 15 = 1$, so

$$A^{-1} = \frac{1}{1} \begin{pmatrix} 4 & -3 \\ -5 & 4 \end{pmatrix} = \begin{pmatrix} 4 & -3 \\ -5 & 4 \end{pmatrix}.$$

Check: $AA^{-1} = \begin{pmatrix} 16-15 & -12+12 \\ 20-20 & -15+16 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$.

Reminder

For $\begin{pmatrix} a & b \\ c & d \end{pmatrix}$ the inverse is $\frac{1}{ad-bc} \begin{pmatrix} d & -b \\ -c & a \end{pmatrix}$ — swap the diagonal, negate the off-diagonal, divide by the determinant.

Exercise 11 Solve $3x + 2y = 12$, $x + 2y = 8$.

Solution. Subtract the second from the first: $2x = 4 \Rightarrow x = 2$. Then $3(2) + 2y = 12 \Rightarrow y = 3$.

Solution $(x, y) = (2, 3)$.

Exercise 12 Solve the 3×3 system.

Solution. Eliminate x : $(\text{eq}_2) - 2(\text{eq}_1)$ gives $-3y - z = -9$, i.e. $3y + z = 9$; and $(\text{eq}_3) - (\text{eq}_1)$ gives $y - 2z = -4$. From the first, $z = 9 - 3y$; substitute: $y - 2(9 - 3y) = -4 \Rightarrow 7y = 14 \Rightarrow y = 2$, then $z = 3$ and $x = 6 - 2 - 3 = 1$. Solution $(x, y, z) = (1, 2, 3)$.

Reminder

Gaussian elimination: use one equation to remove a variable from the others, repeat, then back-substitute. It is the 2×2 trick scaled up.

Exercise 13 $(1, 0, 0), (0, 1, 0), (1, 1, 0)$.

Solution.

(a) Not independent: $(1, 1, 0) = (1, 0, 0) + (0, 1, 0)$.

(b) Rank = 2 : they all lie in the $z = 0$ plane, so they span a plane, not all of $\mathbb{R}^3$.

Reminder

Rank = number of independent directions = dimension of the space the vectors span. A dependent set has rank smaller than its size.

Exercise 14 Three neurons with $\mathbf{n}_3 = 2\mathbf{n}_1 - \mathbf{n}_2$.

Solution.

(a) Not independent — one vector is a fixed combination of the other two.

(b) The effective dimensionality (rank) is 2 .

(c) The population code is *redundant* / *low-dimensional*: although three neurons are recorded, their joint activity lives on a 2-D plane.

Reminder

Real neural populations are often far lower-dimensional than the number of neurons. Finding that low-dimensional structure is exactly what PCA does (next session).

Exercise 15 $\mathbf{A} = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$.

Solution.

(a) $\mathbf{A}(1, 0) = (0, 1)$ and $\mathbf{A}(0, 1) = (-1, 0)$: the axis vectors rotate a quarter turn, so $\mathbf{A}$ is a 90° counter-clockwise rotation.

(b) $\mathbf{A}^2 = \begin{pmatrix} -1 & 0 \\ 0 & -1 \end{pmatrix} = -\mathbf{I}$: applying the rotation twice is a 180° rotation (every vector is reversed).

Reminder

The columns of a matrix are the images of the basis vectors $(1, 0)$ and $(0, 1)$ — read a transformation straight off its columns.